# The Block Time Mirage

*Disentangling Schedule Padding from Operational Reality*

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*Data: BTS Marketing Carrier On-Time Performance (Full Year 2025)*

## Executive Summary

This analysis of 7.7 million flight observations from 2025 evaluates American aviation performance beyond the traditional 15-minute on-time threshold. Data indicates that current reporting frameworks may be providing an incomplete picture of operational quality by comparing carriers using a metric heavily influenced by strategic scheduling buffers, across wildly varying conditions.

71% of U.S. commercial flights in 2025 were scheduled longer than they actually flew. The average buffer built into listed arrival times was around 4.9 minutes per flight. Southwest and Spirit were observed as the most aggressive padders – with 6.7 minutes and 6.2 minutes respectively, directly inflating their positions in the Department of Transportation's official on-time rankings. The metric that passengers use to compare airlines is, to an extent, measuring how conservative each airline schedules its flights.

Breaking down performance into mainline and regional operations reveals what aggregated statistics conceal entirely. American Airlines' regional partners outperform the mainline operator by 3.1 minutes of average arrival delay. United Airlines' regional network underperforms the main carrier by 3.8 minutes. Delta's regional and mainline operations are statistically indistinguishable. Such differences are reflected by the degree to which each marketing carrier governs its regional subsidiaries.

The most extreme case is GoJet Airlines, operating as United Express. GoJet averages 21.3 minutes of arrival delay – nearly six times worse than Republic Airways flying similar routes under the same United brand.

A fixed effects model controlling for origin and destination airport, month, day of week, and time finds that Spirit and United are the best-performing carriers in the dataset, after structural factors are removed; arriving 5.24 and 5.07 minutes earlier than American when conditions are equal. The formerly independent Hawaiian Airlines, who appeared towards the middle of the pack in raw statistics, becomes the worst performer once its favorable island-operating environment is controlled: 11.05 minutes later than American on an equal basis was par for the course for Hawaiian.

Carrier identity alone, however, only explains 3.1% of total delay variance, with the dominating predictors being structural: departing before 8am reduces average departure delays by a factor of six compared to evening flights. The month of September produces delays which are one-ninth the size of July's. Newark Liberty International Airport (EWR) and Chicago O'Hare International Airport (ORD) amplify delays for every carrier that operates through them. The current framework of carrier-level performance ranking obscures these structural nuances.

## 1. Introduction

As someone who's always been intrigued by air travel, and the vibe of an airport for that matter, I've used airports enough to pick up on nuances between the booked airline of choice, and the airline who's operating my flight. The plane visible at the gate wears a regional carrier's livery, deviating from the standard name of the mainline carrier that I had been expecting to see. The ticket confirmation says one name, while the experience delivers a deviation from my original expectations. When the flight is delayed, I find myself asking "why?" far more often than any other question, and more importantly, whether the airline I had entrusted my money with had anything to do with the delay I find myself sitting through.

This experience isn't unique. The American domestic aviation sector operates through a layered system of marketing carriers and operators that most passengers often overlook or brush over. When American Airlines, Delta, or United share their on-time statistics, that data often aggregates the performance of multiple regional airlines under the primary brand identity. A passenger who books Delta based on its reliability when it comes to delays may find themselves on an Endeavor Air or SkyWest aircraft, operating under different conditions than the buyer originally expected.

The official metric used to measure and compare performance between airlines is known as the DOT's 15-minute on-time threshold, though this metric adds layers of confounding that only exacerbate the problem at hand. Airlines usually don't accept this metric as its given; they engineer their schedules to optimize around it and pad their published block times to increase the likelihood that any given flight will clear this threshold, regardless of the actual transit time. Carriers who pad more aggressively will appear to be more reliable in data, even when the operations are more or less identical to other carriers.

This paper examines these distortions using 7.7 million flight records from 2025, with three main findings presented. Firstly, schedule padding is widespread across all airlines and varies significantly by carrier, in such ways that directly blur the on-time metric. Second, regional operator quality can vary dramatically under the same marketing carrier, with the gap between the best and worst operators reaching nearly 18 minutes of average arrival delay. Third, structural factors – particularly the airport of origin and departure time, explain far more of the observed delay than simply the "carrier" operating the aircraft, suggesting that the framework in place: carrier-level performance ranking, is misleading consumers by nature.

An experimental fixed effects model is included as an additional analytical layer, providing the attempt in this analysis to isolate the true carrier effect on delays, after structural controls are applied. These results are presented with appropriate methodological caveats and are primarily intended to demonstrate the directions and approximate magnitudes of carrier-specific effects, rather than precise, causal estimates.

# 2. Background

## 2.1 The DOT On-Time Reporting Framework

The U.S. Department of Transportation defines an on-time arrival as any commercial flight that lands within 15 minutes of its published arrival time, to offer leniency to airlines in the event of trivial matters that only cost a few minutes. This baseline has been the standard since the late 1980s, and it remains the primary metric used by consumers, travel agencies, and airline marketing agencies to compare and contrast carrier performance.

The 15-minute threshold has created an implicit structure to incentivize airlines to publish conservative, padded schedules that have a higher likelihood of clearing that threshold easier, compared to airlines who publish tighter, more “realistic” schedules, regardless of whose aircraft moves more reliably or faster. As the use of data has become more sophisticated in the airline industry, schedule padding has evolved into more of a deliberate strategy rather than a passive buffer.

The consequence of this trend is a metric that blurs scheduling strategy with operational performance. An airline who adds 10 minutes to every published flight will, in proper circumstances, report higher on-time rates than one that publishes accurate schedules, even in cases where both aircraft depart and arrive at the same times. The DOT’s metric doesn’t distinguish between the two aforementioned cases.

To be clear, block buffers are a structural necessity for managing complex networks. Airlines must pad schedules to absorb extended taxi times at sprawling hubs, protect downline passenger connections, and ensure crews do not time out under federal duty regulations.

## 2.2 The Regional Operators

Major U.S. carriers operate significant amounts of their domestic networks through purchase agreements with regional airlines. Under these agreements, a regional carrier, the likes of SkyWest, GoJet, or PSA Airlines operates the aircraft, employs the crew, and conducts the necessary maintenance – while the marketing carrier: American, Delta, United, Alaska – sells the tickets, puts together the schedule, and inherits the responsibility for the passenger experience, with their name representation.

From the perspective of passengers, these distinctions are largely invisible. A ticket booked on United for a United Express flight may have the gate agents, boarding pass, and aircraft livery all carry the United brand, while the operational reality is GoJet Airlines; a separate company with its own standards and operations.

BTS On-Time data captures this distinction through two unique fields: one that shows the selling carrier, and the other showing the operating carrier. When there's a difference between the two, the flight is labeled as a regional operation, performed under what is known as a codeshare agreement. This analysis exploits that distinction to isolate the performance of regional operators from their mainline parents.

## 2.3 Infrastructure Constraints in U.S. Aviation

Not all airports are created equal, and opportunities for on-time performance are never the same across two unique airports. Slot-controlled airports, highly congested facilities, and hubs with higher traffic densities create structural conditions for delays that any carrier operating through them must absorb. A carrier with 30% of their operations at Newark (EWR) will report higher aggregate delays than a carrier who primarily operates through smaller, less crowded, secondary airports, regardless of any operational differences.

EWR has been the punchline of operational inefficiency for years, coming to a head during a staffing crisis within FAA Air Traffic Controllers. In November 2025, 9,098 ATC-attributed cancellations were recorded, largely in part due to the month-long government shutdown that occurred during the period. EWR-exposed carriers took on a disproportionate share of these cancellations. Such an event shows the degree to which infrastructure constraints can overload carrier-level operational differences in the observed data.

# 3. Data and Methodology

## 3.1 Data Source

This analysis observes the BTS Marketing Carrier On-Time Performance dataset, providing flight-level records for U.S. certified air travel carriers who operate aircraft with 10 or more seats. The full year 2025 was observed, spanning January to December and in total, covers 7,736,770 flights.

Key fields in the analysis include: MKT_UNIQUE_CARRIER and OP_UNIQUE_CARRIER (for the marketing versus operating carrier classification), ARR_DELAY and DEP_DELAY (arrival and departure delay in minutes, relative to published schedule), CRS_ELAPSED_TIME and ACTUAL_ELAPSED_TIME (scheduled and actual block times, used for the padding calculations), CANCELLED and CANCELLATION_CODE (for the cancellation analysis), DIVERTED and relevant diversion metrics, delay cause attributions, ORIGIN, MONTH, and DAY_OF_WEEK (structural controls for the experimental fixed effects model).

## 3.2 Key Definitions

A flight is classified as “mainline” when MKT_UNIQUE_CARRIER and OP_UNIQUE_CARRIER are identical. When these fields differ, the flight is classified as a regional operation, which is indicative of a codeshare or capacity purchase agreement.

Schedule padding is calculated as CRS_ELAPSED_TIME minus ACTUAL_ELAPSED_TIME, with positive values indicating that the flight was scheduled for longer than it actually took, which represents a built-in padding buffer in the published timetable. Negative values indicate that the flight ran longer than the scheduled block time.

The DOT on-time threshold of 15 minutes is used throughout the analysis as the standard of delay classification. For percentage calculations, flights that arrive more than 15 minutes late are classified as delayed, consistent with the federal reporting standard. For severe delay events, a 60-minute threshold is used to identify such cases.

Cancelled flights are excluded from delay calculations and are analyzed in a separate manner in the cancellation section. Diverted flights present themselves as an edge case, as their ARR_DELAY field is typically missing, or reflects an irregular outcome. These are flagged, but included in aggregate calculations, as their volume is small enough (0.27% of flights) to have a miniscule impact on averages.

### 3.3 Fixed Effects Model

The fixed effects regression is estimated using the "feols" function from the "fixest" package in R, on the 7,597,495 non-cancelled flights with valid arrival delay data. The model specification is as follows:

$$ARR\_DELAY \sim carrier \mid origin + dest + month + dow + dep\_hour$$

Where carrier is MKT_UNIQUE_CARRIER with American Airlines (AA) is the baseline, origin and "dest" are the three-letter IATA airport code, month is a variable for the calendar month, dow is a variable for day of the week, and dep_hour is a variable for the hour of departure. Standard errors are clustered by origin airport, to account for within-airport correlation in delay statistics.

In total, 364 origin and destination airport fixed effects, 12 month fixed effects, 7 day-of-week fixed effects, and 24 departure-hour fixed effects are absorbed by the model, leaving the carrier coefficients to capture residual delay which can be attributed to carrier identity, after the structural factors are controlled for. The baseline carrier – American Airlines, is excluded from the coefficient output and is instead used as the reference category; all other carrier coefficients represent minutes of arrival delay relative to that of American in equal conditions.

This model is presented as experimental. The low adjusted R-squared value of 0.031 indicates that carrier identity alone only explains around 3.1% of total delay variance, even after controlling for structural constraints, confirming that most delay is driven by factors outside of the control of the carrier. This model's value is in isolating directional carrier effects, and testing the statistical significance of such effects, not in producing precise causal estimates. These results should be interpreted accordingly.

## 4. Finding 1 – The Schedule Padding Problem

Of the 7,597,495 non-cancelled flights observed in the 2025 dataset, 71.3% are scheduled for longer than they actually fly. The average amount of padding across all flights is 4.9 minutes, with a median of 7 minutes amongst these padded flights. This implies that a typical padded flight has a bonus buffer of seven minutes relative to its actual transit time.

This system-wide practice of padding has a direct effect on the on-time statistics. A carrier that adds 7 minutes to every published time will, in average circumstances, convert a sizable number of what would be delays into on-time arrivals. The DOT metric counts such occurrences as genuine performance, which may prove to be misleading.

### 4.1 Padding by Carrier

Schedule padding varies significantly across U.S. carriers in ways that correlate inversely with their reputations for their reliability. Southwest Airlines pads by an average of 6.7 minutes – the most aggressive in the dataset. The recently-shutdown Spirit Airlines, which was commonly associated with unreliability, pads by 6.2 minutes, registering as the second highest. Delta, regarded by many as the pinnacle of U.S. airline operations, pads its flights by 5.4 minutes. Hawaiian Airlines, before merging with Alaska, operated in the most predictable flight environment, padding their flights by only 1 minute on average.

| Carrier | Avg Padding (min) | Median Padding (min) | % of Flights Padded |
|---|---|---|---|
| **Southwest** | 6.7 | 8 | 79.9% |
| **Spirit** | 6.2 | 8 | 73.9% |
| **JetBlue** | 5.5 | 8 | 70.6% |
| **Delta** | 5.4 | 7 | 73.1% |
| **Frontier** | 4.5 | 7 | 69.7% |
| **United** | 4.4 | 7 | 68.8% |
| **American** | 4.1 | 6 | 68.7% |
| **Alaska** | 2.9 | 4 | 62.8% |
| **Allegiant** | 2.3 | 4 | 63.1% |
| **Hawaiian** | 1.0 | 1 | 51.8% |

*Table 1 – Schedule Padding by Carrier (Source: BTS Marketing Carrier On-Time Performance, 2025)*

These padding rankings partially invert the commonly held beliefs amongst the carrier hierarchy. Spirit, who was routinely ranked towards the bottom of the barrel in terms of reliability, padded more aggressively than any legacy carrier. Delta's reputation as a highly efficient product may be partially due to the 5.4-minute average padding. Hawaiian's mediocrity in raw delay stats was reflective of its more honest scheduling, rather than inferior operations.

## 4.2 The Most Padded Routes

Route-level padding reveals specific corridors where airlines make more deliberate decisions to build larger buffers. The United route from San Francisco (SFO) to Honolulu (HNL) registers an average padding of 15.6 minutes, which is the highest in the dataset amongst all routes with more than 1,000 annual flights. The average arrival delay for this route is negative 4.8 minutes, meaning most flights land ahead of their published arrival time by a significant margin. Delta's route from Salt Lake City (SLC) to New York's JFK airport carries 14.8 minutes of padding on average, with an average arrival delay of 1.7 minutes, demonstrating how conservative block times can heavily insulate official on-time records.

Such routes demonstrate the mechanism precisely: larger padding buffers convert what would be ordinary or subpar performance into "excellence". A flight scheduled to take three hours that routinely arrives in two hours and 45 minutes will report as early, regardless of the operations at hand, assuming no external factors cause any setbacks.

## 4.3 Implications for Consumer Reporting

The DOT's on-time metric was designed with the intention of giving passengers a genuine view of the reliability of U.S. airlines. The padding data reveals that the metric is, to an extent, measuring scheduling strategy rather than simple operation quality. Carriers who are more conservative with their padding face a structural disadvantage in headline rankings, relative to the more aggressively-padding carriers, regardless of their actual performance.

A more informative and realistic metric would involve comparing block times to a standardized route benchmark, rather than to each individual carrier's own published schedule, which is highly subject to strategic block inflation. Until such a metric is implemented, official on-time rankings should be interpreted with an understanding: they reflect, in part, how much time an airline has built into its timetable.

## 5. Finding 2 – The Regional Operator Gap

When a passenger books a flight on a legacy U.S. carrier, they may find themselves booking a flight that a separate, regional airline will carry out. The marketing carrier sets their schedule, sells the tickets, and takes the credit, or blame, for the flight experience. However, the crew and operational decisions belong to a different agency that the passenger did not "choose" when choosing to book that flight.

The data from 2025 suggests that this distinction has a massive impact on delay outcomes. The gap between the best and worst regional operators reaches 17.6 minutes of average arrival delay. That difference alone can be the difference in a carrier ranking amongst the best, and one that would rank among the worst carriers.

### 5.1 Mainline vs. Regional Performance by Carrier

Separating each carrier's performance by mainline and regional components reveals patterns that are simply unable to be captured in aggregate statistics.

American Airlines' regional partners: PSA Airlines, Envoy Air, and Republic Airways amongst others, collectively average 10.6 minutes of arrival delay, whereas American's mainline operation averages 13.7 minutes – the regional network is outperforming the mainline by 3.1 minutes despite operating shorter routes, which are, in theory, harder to stay on-time for. American's regional carriers fly routes at an average of 466 miles, compared to 995 miles for the mainline carrier. Shorter routes should produce smaller delays on paper, yet the pattern holds when controlling for distance.

United Airlines represents the opposite, arguably more concerning direction: the mainline averages 7.3 minutes of delay; a respectable benchmark that would place it in the top half of most metrics. United's regional networks, however, averaged 11.1 minutes of delay in 2025, which drags the aggregate output down to 9 minutes.

Delta Air Lines threads the needle: its mainline averages 5.69 minutes of delay, whereas its regional partners average 5.63 minutes. Endeavor Air and select SkyWest flights operating under Delta produce outcomes that are essentially indistinguishable from Delta's own aircraft, reflecting Delta's tighter vertical integration with its regional partners.

| Carrier | Operation | Flights | Avg Delay (min) | Avg Distance (mi) |
|---|---|---|---|---|
| **American** | Mainline | 952,841 | 13.70 | 995 |
| **American** | Regional | 986,835 | 10.60 | 466 |
| **Delta** | Mainline | 1,012,427 | 5.69 | 959 |
| **Delta** | Regional | 588,326 | 5.63 | 441 |
| **United** | Mainline | 786,377 | 7.31 | 1,119 |
| **United** | Regional | 656,729 | 11.10 | 451 |
| **Alaska** | Mainline | 241,927 | 4.80 | 1,376 |
| **Alaska** | Regional | 173,883 | 4.79 | 471 |

*Table 2 – Mainline vs. Regional performance by carrier. Source: BTS Marketing Carrier On-Time Performance (2025)*

### 5.2 The SkyWest Experiment

SkyWest Airlines (OO) represents a highly unique case: it operates its regional flights under contract for four different marketing carriers simultaneously: Alaska, American, Delta, and United. This creates the conditions for a natural experiment of unusual value; the same operator, flying aircraft on rather comparable short-haul routes, and produces outcomes that surprisingly vary in a high amount depending on which carrier's standards are overseeing the operations.

| Marketing Carrier | Flights | Avg Delay (min) | % 15+ min | Avg Distance (mi) |
|---|---|---|---|---|
| **Delta (DL+OO)** | 252,553 | 2.55 | 14.3% | 414 |
| **Alaska (AS+OO)** | 77,647 | 6.47 | 20.8% | 469 |
| **United (UA+OO)** | 338,344 | 12.5 | 22.8% | 450 |
| **American (AA+OO)** | 156,697 | 13.9 | 23.4% | 517 |

*Table 3 – SkyWest Airlines performance by marketing carrier. Routes of comparable average distance. Source: BTS Marketing Carrier On-Time Performance (2025)*

SkyWest under Delta produces an average arrival delay of 2.55 minutes, whereas the same company, flying under American produces 13.9 minutes – a difference of over five times.

Such a difference is harder to justify as simply coming down to the route mix, aircraft, or weather.

Republic Airways (YX) falls under a similar scenario: 3.69 minutes under United, 5.41 minutes under Delta, and 8.88 minutes under American.

### 5.3 The GoJet Case Study

GoJet Airlines (G7) operates exclusively as United Express, which makes it a clear example of a regional operator whose performance is tied to a single marketing carrier's relationship. The aggregate performance for GoJet in 2025 – 21.3 minutes average arrival delay on 72,196 observed flights, makes it the worst-performing regional operator by a substantial margin.

An amount of this poor performance can be attributed to the route-level data: GoJet's worst routes are highly concentrated at EWR. The ORF-EWR route produced an average delay of 57.7 minutes in 2025. TVC-EWR registered at 54.2 minutes. AVL-EWR comes in at 51.9 minutes. Five out of the top 10 worst routes for GoJet terminate at Newark; a pattern that more so reflects the effect of operating into the worst-performing major hub in the dataset.

The monthly trend shows deteriorating operations across the year. GoJet averaged a respectable 10.5 minutes of delay in January, though already the highest of any United regional partner. But this reached 41.9 minutes in December, signaling that operations had broken down to an extent. It is important to recall the government shutdown which greatly impacted the availability of ATCs; where EWR felt the brunt of that impact.

Additionally, GoJet operates routes averaging 371 miles, which is the shortest of any United regional carrier in the dataset. Republic Airways (YX) averages 454 miles and produces 3.69 minutes of average delay. Mesa Airlines (YV) averages 617 miles and produces 4.81 minutes. When controlling for distance, GoJet is seemingly flying the "easier" routes by distance yet is producing delays several times worse than other regional carriers in the family.

GoJet's cancellations, on the other hand, are almost entirely attributable to weather and ATC concerns; carrier-coded cancellations never exceeded five in a single month. This suggests that the airline rarely cancels flights out of operational failure, it often chooses to depart and arrive late to preserve the flight itself. The combination of near-zero carrier cancellations with poor delay performance suggests that a system-wide problem with schedule recovery may be present, rather than simply a capacity management failure.

## 6. Finding 3 – Infrastructure Bottlenecks

Aggregate delay data shows wide variations across origin airports, that dwarfs any potential variation amongst carriers. This should be seen as unsurprising; airports with slot controls, higher levels of congestion, and higher traffic densities impose likelier delay conditions that carriers cannot avoid by operating through them. The data demonstrates the magnitude of this effect and the identity of the airports who are the most culpable of it.

### 6.1 The Newark Gravity Well

Newark (EWR) produces the worst delay results of any major hub in the 2025 dataset. The roundtrip corridor between EWR and Reagan National Airport (DCA) average 28.9 minutes when landing at DCA, and 25.3 minutes when landing at EWR. Those two numbers represent the single worst, and fourth-worst routes in the dataset amongst all routes with more than 2,000 annual flights.

This pattern is by no means carrier specific. Delta, American, United, Frontier, JetBlue, and Spirit all showed their worst route-level performances on EWR corridors. The Atlanta ATL-EWR route, which would normally be a typical hub-to-hub operation, averages 26.0 minutes across 5,395 flights. Chicago O'Hare (ORD) to EWR averages 23.7 minutes. Orlando (MCO) to EWR averages 22.3 minutes. Every airline that reaches Newark is, in some way, penalized by its structural disadvantages.

November 2025 was an extremely troubling month, given the staffing crises during the government shutdown. That month recorded 9,098 NAS/ATC cancellations system-wide, which is more than the combined number of cancellations of the other 11 months. GoJet, with its heavy concentration around EWR, inherited a disproportionate share of these cancellations, on top of its already elevated delay profile thanks to "normal" conditions at EWR being less than ideal.

### 6.2 The O'Hare Effect

Chicago O'Hare International Airport (ORD) functions as the midwestern equivalent of EWR, while also taking on more competition from other carriers. While EWR is primarily a United fortress, all legacy carriers have sizable stakes in ORD. The airport itself ranks as the 11th worst departure airport in the dataset, with an average departure delay of 18.3 minutes, but this effect is most visible in the shorter, regional spoke routes that feed into it.

TVC-ORD (Traverse City to Chicago) averages 25.4 minutes of arrival delay. FSD-ORD (Sioux Falls to Chicago) averages 25.3 minutes. ORD-SBN (Chicago to South Bend) averages 24.6 minutes. These routes, who are mainly operated by smaller, regional contractors flying 50-

seat aircraft into one of the country's most congested airspaces, consistently rank among the worst in the nation, regardless of the identity of the carrier or operator assigned.

The ORD pattern reinforces the central infrastructure finding: smaller regional airports who feed into congested hubs face higher delay outcomes, primarily driven by the hub's constraints rather than by local operational factors. This may cause several regional carriers to have higher average delays.

## 6.3 Seasonal and Temporal Patterns

The monthly data shows two distinct peak periods, separated by a significant September trough. June and July produce the highest average arrival delays in the dataset: 16.1 and 17.3 minutes respectively. This can be largely attributed to summer weather and demand shifts – thunderstorms are far more common during these months and are among the most common reasons for delays, compounded by the increase in travel demand during these months. With warmer weather, and summer breaks for students, travel becomes highly sought-after during June and July.

The average delay falls all the way down to 1.8 minutes in September, the lowest in the data set, reflecting the reduced traffic volume and the end of the peak of thunderstorm season. December, at 13.5 minutes average arrival delay, is notably worse than October (5.2 minutes) and November (6.4 minutes), suggesting that increased holiday traffic and the start of winter weather events contribute to a second peak in delays, that is frequently under-accounted for in planning.

January 2025 reported a high number of cancellations, and the highest rate at that: 3.13%. Of those cancellations, 16,850 out of 18,740 were attributable to weather conditions, primarily due to winter storms across multiple regions. Ironically, the two hubs primarily criticized for their operations; ORD and EWR, both fall in regions highly prone to winter storms. The spike in cancellations did not substantially affect arrival delays for completed flights, as most of the impacted flights were cancelled before an arrival could take place.

The day-of-week data shows a pattern that falls in line with the structure of business and leisure travel demand – Sunday produces the highest arrival delays, with an average of 13.4 minutes, reflecting the higher concentration of travelers returning from weekend trips in time for the start of the next work week. Monday registers second on this list, at 10.2 minutes, driven by business travel peaks at major hubs. Tuesday and Wednesday are the lowest amongst the days of the week, with 4.3 and 5.2 minutes respectively. On these days, demand and congestion are far more manageable.

Regarding the time of departure, rather than simply the day of departure, a highly noticeable fluctuation is spotted. Flights departing between 6am and 8am average four minutes of departure delay. By 5pm, this number rises all the way to 21 minutes. Wait until 7pm, and it reaches over 23 minutes. This reflects the effects of delay compounding across a full day: an aircraft that arrives late in the morning carries into the next flight on that plane, and so on. By evening, the fleet is squarely behind schedule regardless of weather or demand conditions.

## 7. Finding 4 – The Fixed Effects Model (Experimental)

The analyses in Sections 4-6 establish that schedule padding, regional operators, and structural factors all contribute in high amounts to observed delay variation across all carriers. What they cannot establish is the “carrier effect” that remains after these factors are accounted for. This question requires a different approach.

This section presents a fixed effects regression model, derived from the 2025 dataset as an experimental addition to this analysis. It is experimental in the sense that it represents an application of econometric methods to this dataset, with acknowledged limitations. Results are presented as directional estimates of these carrier effects rather than causal claims.

### 7.1 Model Specification and Results

American Airlines serves as the baseline carrier. All other carrier coefficients represent minutes of arrival delay relative to American, in similar structural conditions: the same airport, month, and day of the week.

| Carrier | Coefficient (min) | Standard Error | Significance |
|---|---|---|---|
| **Delta** | -4.64 | 0.511 | *** |
| **Spirit** | -5.24 | 0.680 | *** |
| **United** | -5.07 | 0.546 | *** |
| **Southwest** | -2.21 | 0.729 | ** |
| **Alaska** | -1.13 | 0.592 | Borderline |
| **JetBlue** | -0.85 | 0.793 | Not Significant |
| **Allegiant** | 0.03 | 1.208 | Not Significant |
| **Frontier** | 2.67 | 1.619 | Borderline |
| **Hawaiian** | 11.05 | 1.127 | *** |

*Table 4: Fixed effects model results. Baseline: American Airlines. Clustered standard errors by origin airport. *** p<0.001. N = 7,597,495*

## 7.2 Key Findings

Spirit and United are the two best-performing carriers in the model, arriving 5.24 and 5.07 minutes earlier than American respectively, when conditions are equal. Both coefficients are statistically significant to a high degree. The proximity of the estimates is notable: United markets itself closer to a premium product, while Spirit, who was commonly associated with unreliability, performed close to identically when controlling structural factors. United's regional networks, many of whom operate through EWR, seemingly drag down its aggregate numbers. When controlling for these factors, United appears far stronger.

Delta and Southwest are also significantly better than American after controls, at 4.64 and 2.21 minutes respectively. Alaska is 1.13 minutes better, bordering on statistical significance. The first four carriers are all statistically distinguishable from American.

Allegiant and JetBlue are statistically indifferent from American when removing structural factors. Allegiant's delay numbers, which appeared concerning in the aggregate statistics, are fully explained by the airports and seasons it operates in. When those structural factors are removed, Allegiant and American operate at equal levels. JetBlue's confidence interval removes the significance, despite having a coefficient of nearly one full minute better than American.

Hawaiian provides a counterintuitive result in the model. The raw statistics appeared reasonable: 6.78 minutes of average arrival delays placed it squarely in the middle of the pack, before merging with Alaska. It's worth noting that Hawaiian operated in a highly predictable environment: mainly on Pacific Island routes with minimal concern for weather, and less hub congestion. When the model controls for these advantages, Hawaiian's performance comes out as the worst in the dataset at 11.05 minutes higher delays relative to American, and this result is statistically significant.

## 7.3 The Low R-Squared and Its Meaning

The low adjusted R-squared of 0.031 implies that the pull model, including the fixed effects, explains approximately 3.1% of total variance with arrival delays. This generally reflects the nature of air travel delays: variance is primarily driven by factors outside of the model, such as weather conditions, mechanical issues, and network issues.

The small R-squared value itself is relevant, in the sense that it implies that consumers cannot meaningfully predict delays based on the carrier they're flying, or even structural factors. The unpredictability around flight delays exceeds any possible systematic difference between any carrier.

# 8. Implications

## 8.1 For Passengers

The findings of this analysis provide many conclusions for air travelers making decisions regarding their itineraries. The most impactful decision a passenger can make isn't the carrier they choose to fly on, but rather the time at which they choose to fly. Choosing to depart in the early morning hours – between 6am and 8am, produces departure delays around five times lower than flying during the evening. However, early-morning flying is not as sought after, given the necessity of having to arrive at an airport hours before actual departure. Nonetheless, the time effect dwarfs any carrier-specific differences in the dataset.

Tuesday and Wednesday are consistently the lowest delay days of the week and offer the best chance of a clean travel experience. Sunday ranks as the worst, given the influx of returning passengers from leisure and business trips. The early autumn months, September and October, are the best months for avoiding delays, whereas the winter and summer months are consistently worse off, due to the concentration of demand, and weather uncertainty that airlines cannot account for.

Passengers who book regional flights under a major carrier should be aware that they may not be flying the exact carrier they booked through. The findings in the data suggest that this has significance; the same route under the same brand produces wildly different results depending on which regional operator is assigned on the route.

## 8.2 For Policy

The schedule padding finding raises some questions about the adequacy of the DOT's current on-time reporting metrics. If 71% of flights are routinely scheduled for longer than they take, and if this padding varies by carrier in ways that benefit aggressive padders, the metric is providing viewers with an inherently flawed view of airline performance. A revised metric that revolves around comparing actual block times to a route benchmark rather than each carrier's own schedule, could be more informative and less likely to have its integrity jeopardized.

The infrastructure findings point to higher level of intervention rather than simply regulating the carriers. Newark's delay problems and design flaws have been well-documented for years. Further investments in major hubs to improve infrastructure and staffing concerns would alleviate the high delays, and improve on-time performance for carriers operating through one of New York's major airports.

### 8.3 Limitations

This analysis is limited to a single year of data, which restricts the ability to identify longer trends, comparisons, or the stability of such findings across differing environments. In 2025, the government shutdown produced unusual results, specifically regarding cancellations, and is not fully representative of normal operating conditions.

The fixed effects model only explains 3.1% of delay variance, which greatly limits the confidence in carrier-specific estimates, even when such findings appear to be statistically significant. The model does not control for type of aircraft or passenger load. Any such variable could confound estimates.

Fare data was unavailable for the full year 2025 in BTS' DB1B data, limiting the methods available to connect operational performance to pricing. The relationship between fares and delay performance remains outside the scope of this analysis.

## Conclusion

The official statistics that U.S. air travelers use to evaluate carrier performance are measuring something that appears to be more complicated than the present nature may hint. A combination of genuine performance, schedule inflation, infrastructure quirks, and regional operator contracts that most passengers lack the visibility of. Separating these components paints a different picture of the industry than headline rankings suggest.

After controlling for structural factors, United and the recently-shutdown Spirit were the best performing carriers in the 2025 dataset, whereas the recently merged Hawaiian appeared as the worst carrier once its favorable conditions are accounted for. American Airlines, the representative of the baseline of the fixed effects model, is outperformed by three of the eight major carriers, for which significant results are available. These findings have been found through 7.7 million flight records covering a full calendar year.

The structural findings; that 71% of flights are scheduled to be longer than they actually take, that days in September and Tuesday mornings are the best conditions for delay avoidance, and constraints at EWR and ORD push delay outcomes more than carriers do, suggest that the most useful reform tactic isn't simply "carrier accountability", but rather a reform of the metrics used to report on-time statistics, and further investment into infrastructure to reduce delays.

For travelers who simply want to know when to fly, and who to fly with: book the earliest departure available, on a midweek date, in the early autumn months if the trip allows for flexibility. However, there is no "right" or "wrong" time or place to fly.

# Appendix

## A. Data Sources

BTS Marketing Carrier On-Time Performance: transtats.bts.gov. Full year 2025, January-December. All U.S. certified air carriers operating aircraft with 10 or more seats. Total of 7,736,770 flight records.

## B. Regional Carrier to Parent Airline Mapping

MQ (Envoy Air) → American Airlines

OH (PSA Airlines) → American Airlines

OO (SkyWest Airlines) → Mapped to operating mainline partner per BTS MKT_UNIQUE_CARRIER field. SkyWest operates under American, Delta, United, and Alaska.

YX (Republic Airways) → Mapped to operating mainline partner. Republic operates under American, Delta, and United.

9E (Endeavor Air) → Delta Air Lines

G7 (GoJet Airlines) → United Airlines

C5 (CommutAir) → United Airlines

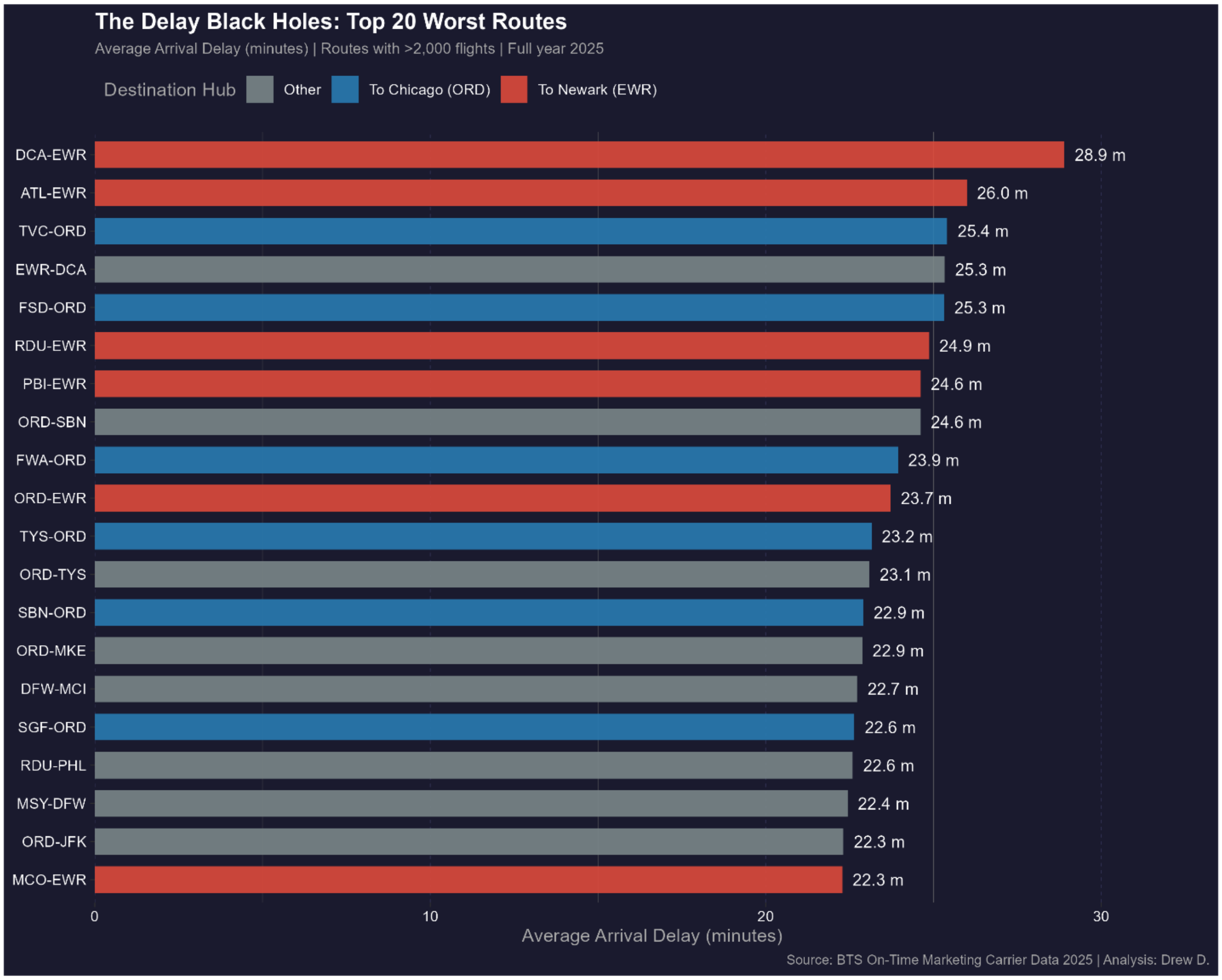
YV (Mesa Airlines) → United Airlines

QX (Horizon Air) → Alaska Airlines

## C. Carrier Distribution Summary

| | MKT_UNIQUE_CARRIER | flights | avg_arr_delay | avg_dep_delay | pct_delayed | pct_cancelled | avg_carrier_delay |
|---|---|---|---|---|---|---|---|
| | <chr> | <int> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> |
| 1 | AS | 422375 | 4.80 | 7.79 | 20.7 | 1.30 | 14.2 |
| 2 | DL | 1626898 | 5.67 | 11.2 | 18.9 | 1.36 | 32.7 |
| 3 | WN | 1391885 | 6.05 | 12.8 | 20.6 | 0.848 | 14.9 |
| 4 | NK | 194515 | 6.60 | 12.9 | 20.9 | 1.50 | 18.7 |
| 5 | HA | 80084 | 6.78 | 7.85 | 16.3 | 0.824 | 32.7 |
| 6 | UA | 1467730 | 9.03 | 13.6 | 21.2 | 1.36 | 23.0 |
| 7 | B6 | 231413 | 11.0 | 16.6 | 25.5 | 1.65 | 20.6 |
| 8 | AA | 1992906 | 12.1 | 16.3 | 23.8 | 2.36 | 25.5 |
| 9 | G4 | 130899 | 13.0 | 15.3 | 24.3 | 0.470 | 25.2 |
| 10 | F9 | 198065 | 14.8 | 19.4 | 27.3 | 1.77 | 23.5 |

## D. Top 20 Most Delayed Routes




## E. Fixed Effects Model Full Output

Model: feols(ARR_DELAY ~ carrier | origin + dest + month + dow + dep_hour, data = fe_data, cluster = ~origin)

```
=== CARRIER EFFECTS RELATIVE TO BASELINE (FINAL MODEL) ===
> print(fe_coefs_final)
       term    estimate carrier_code carrier_name
1 carrierNK -5.24498445           NK       Spirit
2 carrierUA -5.06642311           UA       United
3 carrierDL -4.64260557           DL        Delta
4 carrierWN -2.20789991           WN    Southwest
5 carrierAS -1.13387363           AS       Alaska
6 carrierB6 -0.84666663           B6      JetBlue
7 carrierG4  0.03399469           G4    Allegiant
8 carrierF9  2.67428160           F9     Frontier
9 carrierHA 11.05316523           HA     Hawaiian
```